CHAPTER 1

LEARNING

We call ourselves learners because **memory** is so central to us. For thousands of years, we have tried to understand how we remember and forget, and how a handful of habits can change what stays with us. The study of **learning science** tries to say which habits work. Most of what we read is gone within a week, as the forgetting curve shows.
The strongest remedy is to practise **retrieval**, pulling an idea back out of memory instead of reading it again, a move every chapter in this book returns to.

Memory

Learning science

Retrieval

1.1 Why Retrieval Works

Each time we recall an idea, the path to it gets easier to find the next time. This is why a test is not only a measure of what we know but a way of learning it.
Rereading feels fluent, and that fluency fools us into thinking we know the page.
A quiz breaks the illusion at once: either the answer comes, or it does not.

1 Ebbinghaus measured the curve on himself in the 1880s, with lists of nonsense syllables.
2 A later replication found nearly the same shape.

the page. Good teachers have long known this, and the research now agrees with them.

1.1.1 Testing beats rereading

Testing effect

The **testing effect**, found again and again since the 1900s, is the gain from recalling an idea over reading it again. A test can take many forms:

Free recall

Cued recall

Practice testing

- **free recall** to write down all you remember;
- **cued recall** to answer from a prompt or a question;
- **practice testing** to work through problems without the answers in view.

Each one asks the mind to rebuild the idea, and rebuilding is what makes it last.

1.1.2 Spacing out sessions

Practice spread over days beats the same practice crammed into one night. This **spacing** effect is among the oldest findings in psychology, and among the least used by students.

Spacing effect

Interleaving

Mixing kinds of problems in one session, called **interleaving**, helps for the same reason.